1 Tangent Vectors

1.1 Tangent Space

1.1.1 Geometric Tangent Vectors

Definition 1.1.1

If a is a point of $\mathbb{R}^n$, a map $w : C^\infty(\mathbb{R}^n) \to \mathbb{R}$ is called a ***derivation at a*** if it is linear over $\mathbb{R}$ and satisfies the following product rule:

$$w(fg) = f(a)wg + g(a)wf.$$

Let $T_a\mathbb{R}^n$ denote the set of all derivations of $C^\infty(\mathbb{R}^n)$ at a. Clearly, $T_a\mathbb{R}^n$ is a vector space under the operations

$$(w_1 + w_2)f = w_1 f + w_2 f, \quad (cw)f = c(wf).$$

Proposition 1.1.2

Let $a \in \mathbb{R}^n$.

(a) For each geometric tangent vector $v_a \in \mathbb{R}^n_a$, the map $D_v|_a : C^\infty(\mathbb{R}^n) \to \mathbb{R}$ defined by

$$D_v|_a f = D_v f(a) = \frac{d}{dt}\Big|_{t=0} f(a + tv).$$

is a derivation at a.

(b) The map $v_a \mapsto D_v|_a$ is an isomorphism from $\mathbb{R}^n_a$ onto $T_a\mathbb{R}^n$.

1.1.2 Tangent Vectors on Manifolds

Definition 1.1.3

Let M be a smooth manifold with or without boundary, and let p be a point of M. A linear map $v : C^\infty(M) \to \mathbb{R}$ is called a ***derivation at p*** if it satisfies

$$v(fg) = f(p)vg + g(p)vf \quad \text{for all } f, g \in C^\infty(M). \tag{1.1}$$

The set of all derivations of $C^\infty(M)$ at p, denoted by T_pM, is a vector space called the ***tangent space to M at p***. An element of T_pM is called a ***tangent vector at p***.

Lemma 1.1.4 ▶ Properties of Tangent Vectors on Manifolds

Suppose M is a smooth manifold with or without boundary, $p \in M$, $v \in T_pM$, and $f, g \in C^\infty(M)$.

(a) If f is a constant function, then $vf = 0$.
(b) If $f(p) = g(p) = 0$, then $v(fg) = 0$.

1.2 The Differential of a Smooth Map

1.2.1 Differential on Euclidean Space

Lemma 1.2.1

Let $U \subseteq \mathbb{R}^m, V \subseteq \mathbb{R}^n$ are open sets. $f : U \to V$ is smooth. Take $a \in v$, define a map

$$\begin{aligned} (df)_a &: T_aU \to T_{f(a)}V \\ (df_a)(D_v^a) &\mapsto D_w^{f(a)}, \quad w = J_f(a)v \end{aligned}$$

Then for each $g \in C^\infty(V)$, we have

$$((df_a)(v))(g) = D_v^a(g \circ f)$$

Note.
$(df_a)(v)$ 对应的导子恰好是在点 $f(a)$ 处将 $g \in C^\infty(V)$ 映到 $D_v^a(g \circ f)$ 的导子.

Definition 1.2.2

If M and N are smooth manifolds with or without boundary and F :

$M \to N$ is a smooth map, for each $p \in M$ we define a map

$$dF_p : T_pM \to T_{F(p)}N,$$

called the ***differential of F at p***. Given $v \in T_pM$, we let $dF_p(v)$ be the derivation at $F(p)$ that acts on $f \in C^\infty(N)$ by the rule

$$dF_p(v)(f) = v(f \circ F).$$

Proposition 1.2.3 ▶ Properties of Differentials

Let M, N, and P be smooth manifolds with or without boundary, let $F : M \to N$ and $G : N \to P$ be smooth maps, and let $p \in M$.

(a) $dF_p : T_pM \to T_{F(p)}N$ is linear.
(b) $d(G \circ F)_p = dG_{F(p)} \circ dF_p : T_pM \to T_{G \circ F(p)}P$.
(c) $d(\mathrm{Id}_M)_p = \mathrm{Id}_{T_pM} : T_pM \to T_pM$.
(d) If F is a ***diffeomorphism***, then $dF_p : T_pM \to T_{F(p)}N$ is an isomorphism, and $(dF_p)^{-1} = d(F^{-1})_{F(p)}$.

1.3 Tangent Bundle

Definition 1.3.1

Given a smooth manifold M with or without boundary, we define the ***tangent bundle of*** M, denoted by TM, to be the disjoint union of the tangent spaces at all points of M:

$$TM = \bigsqcup_{p \in M} T_pM.$$

Remark.

- ***projection map*** $\pi : TM \to M$, $\pi(p, v) = p$.

Proposition 1.3.2

For any smooth n-manifold M, then tangent bundle TM has a natural topology and smooth structure that make it into a $2n$-dimensional smooth manifold. With respect to this structure, the projection $\pi : TM \to M$ is smooth.

Proof. Given a chart (U, φ) for M, $\pi^{-1}(U) \subseteq TM$ is the set of all tangent vectors to M at all points of U. Define a map $\tilde{\varphi} : \pi^{-1}(U) \to \mathbb{R}^{2n}$ by

$$\tilde{\varphi}\left(v^i \left.\frac{\partial}{\partial x^i}\right|_p\right) = \left(x^1(p), \cdots, x^n(p), v^1, \cdots, v^n\right)$$

$\square$